

KEN1540 Numerical Methods exam

KEN1540 Numerical Methods 23/24 (period 5-resit semester 2) · 7 exercises · 100.0 points

1 Computer Arithmetic and Algebraic Equations

16.0 points · 6 questions

Consider the problem for finding a solution of the algebraic equation $e^x - 6x - 2 = 0$ in the interval $[0, 4]$.

Text

a Compute one step of Newton's algorithm, starting from the midpoint of the interval.

4.0 points · Open · 3/5 Page

+1 point

Newton: $p_{n+1} = p_n - f(p_n)/f'(p_n)$

+1 point

Derivative: $f'(x) = e^x - 6$

+1 point

Calculation:

$$p_1 = p_0 - f(p_0)/f'(p_0)$$

$$p_1 = 2 - f(2)/f'(2)$$

$$p_1 = 2 - (e^2 - 12 - 2)/(e^2 - 6)$$

$$p_1 = 2 - (-6.6109)/(1.3891)$$

$$p_1 = 6.7593$$

+1 point

Answer:

$$p_1 = 6.7593 \text{ (4dp)}$$

b What happens, and why?

2.0 points · Open · 3/10 Page

+1 point

We leave the interval of interest.

+1 point

Interval too wide, bad starting point, no convergence guarantees for Newton, etc.

c Perform one step of the bisection method to find a smaller bracket for the solution.

3.0 points · Open · 2/5 Page

+1 point

$$f(0) = -1,00 < 0$$

$$f(4) = 28,60 > 0$$

+1 point

$$f(2) = -6.61 < 0$$

+1 point

Hence, $x \in [2, 4]$

d Perform one more step of Newton's method, starting from the midpoint of the interval obtained in part c.

2.0 points · Open · 2/5 Page

+1 point

$$p_1 = p_0 - f(p_0)/f'(p_0)$$

$$p_1 = 3 - f(3)/f'(3)$$

$$p_1 = 3 - (e^3 - 18 - 2)/(e^3 - 6)$$

+1 point

$$p_1 = 2.9939 \text{ (4dp)}$$

e Do you expect your new approximation to be within 0.1 of the true solution? Give a reason for your answer.

2.0 points · Open · 2/5 Page

+1 point

Yes

+1 point

Because $|p_1 - p| \leq |p_1 - p_0| = |2.9939 - 3| = 6.1e-3$

f Write a Matlab statement which performs a single step of Newton's method for a function f , starting from point x .

3.0 points · Open · 1/2 Page

+1 point

`f = @(x) exp(x) - 6*x - 2;`

+1 point

`df = @(x) exp(x) - 6;`

+1 point

`x = x - f(x)/df(x);`

+3 points

Alternative: `x = x - (exp(x)-6*x-2)/(exp(x)-6);`

2 Differential Equations

20.0 points · 3 questions

Consider the problem of finding the value of $y(3)$ for the initial value problem

$$\frac{dy}{dt} = \frac{1}{t-1} - y^2, \quad y(2) = 1.4.$$

Text

a Solve the problem using Ralston's method with a step size of $h = 0.5$.

10.0 points · Open · 1 2/5 Page

+1 point

Ralston's method $y_{n+1} = y_n + \frac{1}{4}h(f(t_n, y_n) + 3f(t_n + \frac{2}{3}h, y_n + \frac{2}{3}hf(t_n, y_n)))$

OR

$k_1 = hf(t, y); k_2 = hf(t + \frac{2}{3}h, y + \frac{2}{3}k_1); y' = y + \frac{1}{4}(k_1 + 3k_2).$

+1 point

$t_0 = 2$ and $y_0 = 1.4000$.

+1 point

$f(t_0, y_0) = f(2.0, 1.4000) = -0.9600$ and

$f(t_0 + \frac{2}{3}h, y_0 + \frac{2}{3}hf(t_0, y_0)) = f(2.\dot{3}, 1.0800) = -0.4164$.

Note $k_{0,1} = -0.4800$ and $k_{0,2} = 0.2082$.

+1 point

$t_0 + \frac{2}{3}h = 2.\dot{3}$ and $t_1 = t_0 + h = 2.5$.

+1 point

$y_0 + \frac{2}{3}hf(t_0, y_0) = 1.0800$.

+1 point

$y_1 = 1.4000 + \frac{1}{4} \times 0.5 \times (-0.9600 + 3 \times -0.4164) = 1.1239$.

+1 point

$f(t_1, y_1) = f(2.5, 1.1239) = -0.5964$ and

$f(t_1 + \frac{2}{3}h, y_1 + \frac{2}{3}hf(t_1, y_1)) = f(2.8\dot{3}, 0.9251) = -0.3103$;

Note $k_{1,1} = -0.2982$ and $k_{1,2} = -0.1551$.

+1 point

$t_1 + \frac{2}{3}h = 2.8\dot{3}$ and $y_1 + \frac{2}{3}hf(t_1, y_1) = 0.9251$.

+1 point

$y_2 = 1.1239 + \frac{1}{4} \times 0.5 \times (-0.5964 + 3 \times -0.3103) = 0.9329$.

+1 point

$t_2 = t_1 + h = 3.0$, so $y(3.0) \approx y_2 = 0.9329$.

-1 points

Too little precision (<3sf) at y_1 ; at least one correct calculation.

-1 points

Much too little precision (<2dp) throughout; at least 3 points in computation.

b How would you expect the error to change if you were to take a step-size of $h = 0.2$?

2.0 points · Open · 2/5 Page

+1 point

Method is 2nd-order, so error $\sim h^2$.

+1 point

Expect error to decrease with a factor roughly $(0.2/0.5)^{d=2} = 0.16$ (decrease by factor 6.25 times).

c Write a Matlab function `ode_ralston` to solve a general differential equation $dy/dt = f(t, y)$ over the time interval $[t_{\text{init}}, t_{\text{final}}]$ with initial value y_{init} using n steps of Ralston's method.

8.0 points · Open · 4/5 Page

0 points

```
function ys = ode_ralston(f, t_init_final, y_init, n)
    t_init = t_init_final(1); t_final = t_init_final(2);
    t = t_init; y = y_init;
    h = (t_final - t_init)/n;
    ys = NaN(1,n+1); ys(1) = y;
    for i=1:n,
        k1 = h * f(t,y); k2 = h * f(t+2/3*h,y+2/3*k1); k = (k1+3*k2)/4;
        t = t+h; y = y+k;
        % fty=f(t,y); y = y + h/4*(fty+3*f(t+2/3*h,y+2/3*h*fty)); t=t+h;
        ys(n+1) = y;
    end
end
```

+1 point

Function signature `function r = ode_xxx(a,b,...)`

+1 point

Function arguments `f`, `t_init`, `t_final`, `y_init` and `n`.

+1 point

Initialise `h`, `t` and `y`.

+1 point

Loop for n steps. Update $t=t+h$ each step.

+1 point

Output array of all y -values. Alternatively, output only final y value.

+1 point

Correct syntax in formulae for new y .

+1 point

Correct formula for new y in reasonable syntax.

+1 point

Only compute $f(t, y)$ once per step.

3 Polynomial Interpolation

16.0 points · 4 questions

Consider the data

x_i	2.0	1.5	1.0	3.0
$y_i = f(x_i)$	0.40	1.00	1.82	0.03

Text

a Compute the divided differences $f[x_i, \dots, x_j]$ for $0 \leq i \leq j \leq 3$. You may assume $f[x_0, x_1] = -1.200$ and $f[x_1, x_2] = -1.640$.

6.0 points · Open · 4/5 Page

+1 point

Finite difference formula.

+1 point

$f[x_2, x_3] = -0.8950$

+1 point

$f[x_0, x_1, x_2] = 0.4400$

+1 point

$f[x_1, x_2, x_3] = 0.4967$

+1 point

$f[x_0, x_1, x_2, x_3] = 0.0567$

+1 point

Correct use of given $f[x_0, x_1]$ and $f[x_1, x_2]$.

b Write down the nested form of the cubic polynomial p_3 interpolating the given data.

2.0 points · Open · 3/10 Page

+1 point

nested form

+1 point

$p_3(x) = 0.4000 + (x-2.0)(-1.2000+(x-1.5)(0.4400+(x-1.0)0.0567))$

c Estimate the value of y when $x = 2.5$, showing intermediate steps in your computations.

3.0 points · Open · 1/2 Page

+1 point

computation

+1 point

correct answer: $p(2.5) = 0.0625$ (4dp)

+1 point

correct rounding: $p(2.5) = 0.06$ (2dp)

d Write a Matlab function to evaluate a polynomial of degree n given in nested form, given the interpolation points x_i and coefficients a_i .

5.0 points · Open · 4/5 Page

0 points

function $z = \text{eval}(a,x,w)$

$n = \text{length}(a);$

$z = a(\text{end});$

for $k = n-1:-1:1$

$z = a(k) + (w-x(k))*z;$

end

end

+1 point

function signature

+1 point

initialization z

+1 point

for loop

+1 point

recursive expression: idea

+1 point

recursive expression: correct

4 Integration & Differentiation

16.0 points · 4 questions

Consider the problem of computing to an accuracy of 0.01 the integral

$$I = \int_{0.0}^{1.2} f(x) dx \text{ for } f(x) = \cos(x^2).$$

Text

a Use the trapezoid rule with $n = 4$ subdivisions to estimate the value of I .

5.0 points · Open · 4/5 Page

+1 point

$$T_4 = h\left(\frac{1}{2}f(x_0) + \sum_{i=1}^3 f(x_i) + \frac{1}{2}f(x_4)\right).$$

+1 point

Step $h = (b - a)/4 = (1.2 - 0.0)/4 = 0.3$ and partition points $x_i = (0.0, 0.3, 0.6, 0.9, 1.2)$.

+1 point

Function values $y_i = f(x_i) = (1.0000, 0.9960, 0.9359, 0.6895, 0.1304)$.

+1 point

$$T_4 = h(y_0/2 + y_1 + y_2 + y_3 + y_4/2) = 0.3 \times \left(\frac{1}{2} \times 1.0000 + 0.9960 + 0.9359 + 0.6895 + \frac{1}{2} \times 0.1304\right).$$

+1 point

Trapezoid approximation $T_4 = 0.3 \times 3.1866 = 0.9560$.

-1 points

Using less than 2dp, and at least one correct computation.

b Use the error estimate for the trapezoid rule to estimate the error over the sub-intervals $[0.0, 0.6]$ and $[0.6, 1.2]$. Do you expect the total error to be less than the desired accuracy bound?

5.0 points · Open · 7/10 Page

+1 point

Error estimate over $[p, q]$ is $\frac{1}{12}(q - p)|f(p) - 2f(\frac{p+q}{2}) + f(q)|$
or $\frac{1}{3}|T_2(f, [p, q]) - T_1(f, [p, q])|$.

May give error bound $\frac{1}{24}(q - p)h^2 \max_{\xi \in [p, q]} |f''(\xi)|$.

+1 point

Over $[0.0, 0.6]$, have $E_0 = \frac{1}{12} \times (0.6 - 0.0) \times |1.0000 - 2 \times 0.9960 + 0.9359|$
or $T_1 = 0.5808$ and $T_2 = 0.5892$.

+1 point

$E_0 = 0.05 \times |-0.0560| = 0.0028 = 2.8 \times 10^{-3}$.

+1 point

Over $[0.6, 1.2]$, have $E_1 = \frac{1}{12} \times (1.2 - 0.6) \times |0.9359 - 2 \times 0.6895 + 0.1304| = 0.05 \times |-0.3127| = 0.0156 \approx 1.6 \times 10^{-2}$.

Or $T_1 = 0.3199, T_2 = 0.3668$.

+1 point

Total error $E_0 + E_1 = 0.018 \dots \leq 0.019$, not less than error bound.

c Explain how to use the adaptive trapezoid method to obtain a better estimate to the integral. (You do not have to perform the computation.

3.0 points · Open · 3/5 Page

+1 point

Recursively subdivide one of the intervals.

+1 point

Expect error on subdivided interval to be smaller (multiplied by $1/4$).

or

Continue until total error is small.

+1 point

Choose the subinterval with the largest error estimate, (here $[0.6, 1.2]$).

d Write Matlab code to compute the integral using a builtin Matlab function for integration.

3.0 points · Open · 1/2 Page

+1 point

Define $f=@(x)\cos(x.^2)$.

+1 point

Use `integral` function.

+1 point

Call with arguments (f,a,b).

5 Least-Squares Approximation

16.0 points · 5 questions

Consider the data:

x_i	0	$\frac{1}{4}\pi$	$\frac{1}{2}\pi$	$\frac{3}{4}\pi$	π	$\frac{5}{4}\pi$	$\frac{3}{2}\pi$	$\frac{7}{4}\pi$	2π
$f(x_i)$	0.550	0.300	-0.033	0.300	0.800	0.300	-0.033	0.300	0.550

Text

a Compute the coefficients a_0, a_1, a_2 of the discrete Fourier transform.

6.0 points · Open · 4/5 Page

+1 point

$$a_k = \left(\sum_{i=0}^7 y_i \cos(kx_i) \right) / 4.$$

+1 point

$$a_0 = \left(\sum_{i=0}^7 y_i \right) / 4 = (0.550 + 0.300 - 0.033 + 0.300 + 0.800 + 0.300 - 0.033 + 0.300) / 4 = 2.484 / 4 = 0.621.$$

+1 point

Using $\cos(\pi/4) = 1/\sqrt{2} = 0.707$ find

$$a_1 = \left(\sum_{i=0}^7 y_i \cos(x_i) \right) / 4 = (0.550 \times 1 + 0.300 \times 0.707 - 0.033 \times 0 + 0.300 \times -0.707 + 0.800 \times -1 + 0.300 \times -0.707 - 0.033 \times 0 + 0.300 \times 0.707) / 4.$$

+1 point

$$a_1 = (0.550 + 0 - 0.800 + 0) + (0.300 - 0.300 - 0.300 + 0.300) \times 0.707 = -0.25 / 4 = -0.0625.$$

or

$$a_1 = (0.550 + 0.212 + 0 - 0.212 - 0.800 - 0.212 + 0 + 0.212) / 4 = -0.25 / 4 = -0.0625.$$

+1 point

$$a_2 = \left(\sum_{i=0}^7 y_i \cos(2x_i) \right) / 4 = (0.550 \times 1 + 0.300 \times 0 - 0.033 \times -1 + 0.300 \times 0 + 0.800 \times 1 + 0.300 \times 0 - 0.033 \times -1 + 0.300 \times 0) / 4.$$

+1 point

$$a_2 = (0.550 + 0.033 + 0.800 + 0.033) / 4 = 1.416 / 4 = 0.354.$$

b Assuming the coefficients b_i equal 0 for all i , write down the second Fourier approximation s_2 to f .

2.0 points · Open · 3/10 Page

+1 point

$$s_2(x) = a_0/2 + \sum_{k=1}^2 a_k \cos(kx).$$

+1 point

$$s_2(x) = 0.3105 - 0.0625 \cos(x) + 0.354 \cos(2x).$$

c Evaluate $s_2(\pi/3)$

2.0 points · Open · 2/5 Page

+1 point

$\cos(\pi/3) = 0.5$ and $\cos(2\pi/3) = -0.5$.

+1 point

$s_2(\pi/3) = 0.3105 - 0.0625 \times 0.5 + 0.353 \times -0.5 = 0.3105 - 0.03125 - 0.1765 = 0.10225 = 0.102$ (3dp).

-1 points

More than 3dp in answer.

d What is the root-mean-square value $\sqrt{\frac{1}{2\pi} \int_0^{2\pi} s_2(x)^2 dx}$?

3.0 points · Open · 1/2 Page

+1 point

$\int_0^{2\pi} s_2(x)^2 dx = \pi(a_0^2/2 + a_1^2 + a_2^2)$ so obtain $p = \sqrt{\frac{1}{2}(a_0^2/2 + a_1^2 + a_2^2)}$.

+1 point

$p = \sqrt{\frac{1}{2}(0.621^2/2 + (-0.0625)^2 + 0.354^2)} = \sqrt{\frac{1}{2}(0.3856/2 + 0.0039 + 0.1253)}$.

+1 point

$P = \sqrt{\frac{1}{2} \times 0.3220} = \sqrt{0.1610} = 0.401$.

e Write Matlab code to compute the Fourier coefficient a_2 for n equally-spaced data points in $[0, 2\pi)$.

3.0 points · Open · 3/5 Page

+1 point

`x=(0:n-1)/n*pi`

+1 point

`z=y.*cos(2*x)`

+1 point

`a2=sum(z)/(n/2)`

-1 points

Syntax errors, e.g. not using `*` for multiplication.

6 Linear Algebra

16.0 points · 3 questions

Let A be the matrix

$$\begin{pmatrix} 7 & 5 \\ 1 & 4 \end{pmatrix}$$

Text

a Apply one step of the QR-method to find a matrix B with the same eigenvalues as A .

9.0 points · Open · 1 1/5 Page

+1 point

$$v_1 = x_1 = [7; 1]$$

+1 point

$$v_2 = x_2 - (x_2 \text{ dot } v_1) / (v_1 \text{ dot } v_1) v_1$$

+1 point

$$v_2 = \dots = [-0.4600; 3.2200] \text{ (computation)}$$

+1 point

$$r_{11} = \|v_1\| = \text{sqrt}(50) = 7.0711$$

$$r_{22} = \|v_2\| = 3.2527$$

+1 point

$$v_1 = v_1 / r_{11} = [0.9899; 0.1414]$$

$$v_2 = v_2 / r_{22} = [-0.1414; 0.9899]$$

+1 point

$$r_{12} = x_2 \text{ dot } u_1 = 5.5151$$

+1 point

$$r_{21} = x_1 \text{ dot } u_2 = 0.0000$$

+1 point

$$A = QR = [0.9899 \ -0.1414; 0.1414 \ 0.9899] * [7.0711 \ 5.5151; 0 \ 3.2527]$$

+1 point

$$A_1 = R * Q = [7.7795 \ 4.4595; 0.4599 \ 3.2198]$$

b Apply the Gershgorin circle theorem to B and on B^T to find bounds on the eigenvalues of A .

4.0 points · Open · 1/2 Page

+2 points

Gershgorin circle theorem

+1 point

Applied to B:

$3.18 \leq \lambda_1 \leq 12.38$

$2.76 \leq \lambda_2 \leq 3.68$

+1 point

Applied to B^T :

$7.32 \leq \lambda_1 \leq 8.24$

$-1.38 \leq \lambda_2 \leq 7.82$

+1 point

Tightest bound for λ_2 from B and for λ_1 from B^T .

c Write code to perform one step of the QR-method on the matrix A in Matlab. You may use a builtin command for performing a QR-factorisation.

3.0 points · Open · 3/5 Page

+1 point

$A = \begin{bmatrix} 7 & 5 \\ 1 & 4 \end{bmatrix};$

+1 point

$[Q,R] = \text{qr}(A);$

+1 point

$B = R*Q;$

7 Extra Paper

0.0 points · 1 question

Open · 2 9/10 Page